

Event Density — The Tiered-Claims Ledger

Event Density is a unified framework of physics. It is based on 13 Primitives, 10 Constants, and 15 Postulates. From these inputs, Event Density reproduces the form and mechanism of over 150 key findings in 12 separate domains of physics.

The tables below outline Event Density from its Theory to its Predictions to its Assertions and Open questions; all honestly tiered.

This ledger leads with Predictions, exactly where Event Density can be broken, because consilience may be compelling, but it is testing that gains acceptance.

Predictions (the bets — where it can be killed)

Domain	Paper	Claim	Tier	Falsifier / kill-condition
Gravity	031	$a_0(z) = cH(z)/(2\pi)$ evolves factor-for-factor with H	Prediction	departs from $cH(z)/2\pi$. Only 031 §8.8 is data-confronted: MUSE-DARK III (A&A 2026) ~30sigma evolution, ~8% local, edge of H(z); $\alpha=1$ the decisive open number. Do NOT over-bank (direction won, exact rate tense)
Gravity	029	$a_0 = cH_0/(2\pi) \approx 1.08 \times 10^{-10} \text{ m/s}^2$, local MOND transition acceleration (the redshift evolution 1.15 is the forward weapon)	Derived	a_0 tracks H_0 wrong as the Hubble tension resolves; wide binaries at a_0 obey plain Newton
Gravity	GR-IV	α_1, α_2 preferred-frame (the keystone kill-switch)	Prediction	ANY nonzero α_2 kills it ($\alpha_2 = 0$ EXACT, structural); $\alpha_1 > \sim 1e-5$ kills it ($\alpha_1 < \sim 4e-93$ safe). Passed favorably
Gravity	OffsetVelocity/117	Offset-velocity knee law: flat-knee-line-ceiling	Prediction	ΛCDM velocity-independent scatter, or MOND-EFE smooth roll-off, refutes. Only the SHAPE is ingredient-robust; the knee location rests on a flagged $\nu \sim 0.70$ (3D O(3)) assumption
Cosmology	DF2/DF4	Group-embedded low-accel galaxies show a KNEE-like suppression sharper than MOND-EFE; void UDGs full enhancement	Prediction	Preds 1-4: environmental independence (ΛCDM) or smooth MOND-EFE roll-off; knee sharper than MOND (Pred-4 uniquely ED, below current precision)
Dark-sector	DarkSector	Dark matter (relic) is WARM (~0.2-0.7 keV), not a cold WIMP	Prediction	F1: a cold heavy WIMP detection, or perfectly-cold DM, refutes the warm relic. LIVE RISK: mechanism prefers cold GeV-Planck (6-26 OOM too heavy)
Dark-sector	DarkSector	Galaxies contain NO dark-matter halo at all	Prediction	F2: DM substance found INSIDE galaxies (halo, CDM cusps, scattered mass) refutes the horizon-only galactic account
Gravity	038.6	Activity-dependent weak lensing $\sim \text{grad } \mu ^2$	Prediction	activity-INDEPENDENT lensing refutes. PROVISIONAL — $\alpha_{\text{act}}, A(x)$, the $ \text{grad } \mu ^2$ -vs-feedback discriminator observable all OPEN
Cosmology	Cos-05	Dark energy strictly $w=-1$, no time-variation (vs quintessence/phantom)	Prediction	F1/F4: clean phantom $w < -1$ over wide z , or $w\text{-dot} \neq 0$, or $\Lambda\text{-dot} \neq 0$. Distinctive $w=-1$ null; ties 22-Ways $w=-1$
Dynamics	Dyn-02	EoS confined to $\{-1, 0, 1/3\} + \text{composites}$ (no clean phantom $w < -1$, no stiff $w=1$)	Prediction	a clean phantom $w < -1$, or a stiff $w=1$, outside the substrate-admitted $\{-1, 0, 1/3\} + \text{composites}$ set refutes it. NEAR-distinctive, but sits BEHIND the Q1/Q2 OPEN (currently inherited, not derived); ties to Cos-05 $w=-1$
Q-compute	056	Class-A wall EXISTS as sharp + architecture-independent (the shape is the claim, not the number)	Prediction	no sharp architecture-independent wall (smooth environment/architecture-DEPENDENT decoherence) refutes it. Standard decoherence-only + CSL predict no such wall. NUMBER 140-250 kDa is PRESSURED (> 170 kDa interference already shown, Nature 2026, still QM-consistent) + the second-harmonic leg CONFOUNDED (KDTL optics); the SHAPE is the claim -- do NOT over-bank the number
Q-compute	058b	Class-C plateau, RE-SCOPED to broadcast-type redundancy (repetition / GHZ / cat), NOT surface-code distance	Prediction	surface-code clean suppression (Willow $\Lambda \sim 2$) is consistent-BY-CONSTRUCTION, NOT a passed test. Live pair = repetition-code floor persistence (Google $\sim 1e-10$ above $d \sim 15$) + cat-state width ceiling; a floor surviving burst-mitigation at cross-platform-consistent content = the ED signature. Standard FT predicts error $\rightarrow 0$
Substrate-evaluation	CleanSubstrateVector	Parity violation is a spontaneous, contingent orientation correlated with the matter/antimatter sign	Prediction	a demonstrated LAW-LEVEL (non-spontaneous, non-contingent, fixed) origin of the weak handedness refutes it. Distinctive vs the SM's law-level fixed handedness
predictions	ED-09.5	Slow envelope modulation at bifurcation, $f_v \approx 1.1 \gamma_{\text{dec}}$ (the Q/π form), tested across 10+ orders of magnitude in decoherence rate	Prediction	No envelope detected at the predicted frequency band (~11–110 Hz) in any of 3+ uncorrelated datasets (optomechanics + FRAP residuals)
qft	MS-II	No parity-violating abelian U(1) force; the abelian coupling is chirality-blind by construction	Prediction	Discovery of a chiral (parity-violating) U(1) gauge force
qft	MS-II	No stable fundamental gauge group larger than SU(3); stable channel families are exactly {1..d} with $d=3$	Prediction	Discovery of a stable fundamental SU($N \geq 4$) sector

Theory (the 38 inputs: 13 primitives, 10 constants, 15 postulates)

Domain	Paper	Claim	Tier	Falsifier / kill-condition
Primitives	P01	Existence layer (the foundational relational existence; below it, no further substrate structure)	Primitive	
Primitives	P02	Participation as the primitive relation (chains participate — derived composite structures)	Primitive	
Primitives	P03	Channel + locus indexing; spatial homogeneity (bundles three commitments, splittable)	Primitive	
Primitives	P04	Bandwidth — a non-negative additive scalar	Primitive	
Primitives	P05	Polarity-transport between adjacent loci	Primitive	
Primitives	P06	Spatial dimension $D = 3+1$	Primitive	
Primitives	P07	Channel structure as an ontological primitive	Primitive	
Primitives	P08	Substrate scale I_{ED} (= L_P empirically)	Primitive	
Primitives	P09	$U(1)$ -valued polarity	Primitive	
Primitives	P10	Rule-type primitive (licenses V5 as a P10 kernel rule-type)	Primitive	
Primitives	P11	Commitment — irreversible, with environmental phase-randomization (THE arrow primitive)	Primitive	
Primitives	P12	Stability landscape $\Sigma = Coh - Str - Grad$	Primitive	
Primitives	P13	Time homogeneity; primitive event-discreteness (NOT 'the tick')	Primitive	
Constants-inherited	Paper_012 (P-RB-1)	c — the rate of becoming (tick-to-hop conversion)	Constant	
Constants-inherited	Paper_027	$G = c^3 I_P^2 / \hbar$ — Newton's constant	Constant*	
Constants-inherited	Paper_029	$a_0 = \hbar c / (2\pi)$ — the MOND transition scale	Constant*	
Constants-inherited	Paper_038_5	Λ — the cosmological constant	Constant*	
Constants-inherited	Paper_087	$\hbar$ — the action of one commitment (the grain of memory)	Constant	
Constants-inherited	Paper_087 (P08) / 089	$L_P = I_{ED}$ — the grain (size of one hop)	Constant	
Constants-inherited	cosmology	H_0 — the Hubble rate	Constant*	
Constants-inherited	matter sector	Particle masses	Constant	
Constants-inherited	matter sector	Gauge couplings g	Constant	
Constants-inherited	matter sector	$\alpha \sim 1/137$ (fine-structure); mass ratios	Constant	
Cosmology	SCBU	a_0 & χ_i canonical FORCED to share one substrate-cosmology boundary origin	Postulated	
Entanglement	063	E-1 factorizable $\Psi^{AB} = \Psi^A \times \Psi^B$ & the tensor-product structure	Postulated	
Entanglement	064	V5-bilocal has Schmidt rank > 1 generically	Postulated	
Entanglement	065	Monogamy budget cap; measure saturates the budget	Postulated	
Entanglement	069	ED sits at Tsirelson; PR-boxes excluded	Postulated	
Gravity	035	Six acoustic-metric guardrails (C1-C6)	Postulated	
Gravity	QuadraticStrain	P-Quadratic-Strain (Model C)	Postulated	
QFT	019	P-YM-Action-Coarse-Graining	Postulated	
QFT	020	P-OS-Reflection-Positivity (OS3)	Postulated	

Domain	Paper	Claim	Tier	Falsifier / kill-condition
QFT	021	P-Profile-Rescaling (YM continuum survival)	Postulated	
QM-kinematics	002	Tensor-product composition $\Psi^{AB} = \Psi^A \times \Psi^B$	Postulated	
QM-kinematics	012	P-RB-1 local rate of becoming; substrate-c constancy	Postulated	
Relativistic-QM	114	P-Gauge (paper-specific postulate)	Postulated	
Substrate-evaluation	QuantumDarwinism	P-QD-LiveWeight (uncommitted branch-basis correlation consumes pairwise V5 weight)	Postulated	
Substrate-evaluation	V5AttractiveSign/RelationalTick/MassWithoutMass	V5 existence as a cross-chain kernel	Postulated	

Claims (~159 derivations and groundings across ~12 domains)

Domain	Paper	Claim	Tier	Falsifier / kill-condition
Black-hole	039	Horizon = $b \rightarrow 0$ decoupling surface (Gamma_cross collapse)	D-via-I / Form-forced	
Black-hole	040	Trans-Planckian resolution (modes terminate at wc)	D-via-I / Form-forced	
Black-hole	041	Planck-mass remnant $M^* = c^* l_P$ (Scenario C)	D-via-I / Form-forced	
Black-hole	042	No-singularity (C_{cum} bounded, $r \geq l_P$)	D-via-I / Form-forced	
Black-hole	043	Area-law FORM $S \propto A$	D-via-I / Form-forced	
Black-hole	044	BHPT scattering / Regge-Wheeler form	D-via-I / Form-forced	
Black-hole	045	Helicity preserved (Schwarzschild) / amplified (Kerr)	D-via-I / Form-forced	
Black-hole	046	Kerr twist $g_{t\phi}$	D-via-I / Form-forced	
Black-hole	047	Hawking temperature $T_H = \kappa/2\pi$ via V5 KMS	D-via-I / Form-forced	
Black-hole	047	Non-thermal Hawking correction $[1 - c_1(w/wc)^2]$	D-via-I / Form-forced	
Black-hole	048	Scenario C forced (A, B excluded)	D-via-I / Form-forced	
Black-hole	049	PBH relic abundance $\Omega_{\text{PBH}} \propto n_{\text{PBH}} M^*$	D-via-I / Form-forced	
Black-hole	050	Page-curve rise + turnover + decline	D-via-I / Form-forced	
Dark-sector	RelicLagrangian	Relic $\rho \propto a^{-3}$ from the P11 commitment number	Derived	
Dark-sector	RelicLagrangian	Sector is TWO-COMPONENT, not one substance (L_{self} coarse-grains CANONICAL)	Derived	
Dark-sector	RelicLagrangian	Pressureless-dust limit $w \sim 0$ (fast-oscillating massive scalar)	Derived	
Entanglement	064	Schmidt rank $> 1 \Rightarrow$ non-factorizable	Derived	
Entanglement	065	Monogamy (CKW-form inequality)	Derived	
Entanglement	066	No-signaling $p(a x,y) = p(a x)$, over-determined by 3 locks	Derived	
Entanglement	068	von Neumann entropy form $S = -\text{Tr } \rho \log \rho$	Derived	
Foundations	089	Theorem N1 — V1 finite-width (G bounded, decaying, scale l_{ED} ; delta-function and inf-width limits excluded)	Derived	
Foundations	089/093	Theorem T18 — V1 strictly retarded support (the kernel-level arrow); advanced-V1 refuted by P11, symmetric-V1 non-constructible	Derived	
Foundations	089/093	Wheeler-Feynman engagement: symmetric electrodynamics is mathematically consistent but substrate-operationally non-constructible in ED	Derived	
Foundations	090/094	V5 inherits retardation from V1 (via T18); primitive-forward-causality $\rightarrow$ phenomenological arrows inherit under DCGT	Derived	
Gravity	025	Holographic bound $N(R) \leq 4\pi R^2/l_{\text{ED}}^2$ (area law as edge-count)	Derived	
Gravity	026	Newton $a_N = GM/R^2$ from the cumulative-strain reading	Derived	
Gravity	032.5	Free chains follow acoustic-metric geodesics	Derived	
Gravity	034/036	Deep-MOND $a = \sqrt{a_N a_0}$; flat-background AQUAL field eq	Derived	
Gravity	037	a_0 continuum-limit invariant (does not RG-run)	Derived	
Gravity	GR-I	Temporal lapse $N^2 \sim b \rightarrow$ weak-field Einstein metric (Schwarzschild branch, not Nordstrom)	Derived	
Gravity	GR-I	Emergent $g \sim 1/b$ + factor-of-two light bending	Derived	
Gravity	GR-II	Field-equation form $G + \Lambda g = \kappa T$	Derived	
Gravity	GR-II	Matter conservation $\text{div } T = 0 \Leftrightarrow$ geodesic motion	Derived	
Gravity	GR-II	ED gravity is KHRONOMETRIC (2 tensor + 1 khronon); gauge = foliation-preserving diffeos	Derived	
Gravity	GR-II	$c_T = c$ structurally (GW170817 as identity); universal (not differential) Lorentz violation	Derived	
Gravity	GR-III	$\alpha=1$ (P-Commitment-Linear) FORCED; Nordstrom excluded	Derived	

Domain	Paper	Claim	Tier	Falsifier / kill-condition
Gravity	GR-III	Khronon scalar speed $c_s = c$	Derived	
Gravity	GR-IV	$\alpha_2 = 0$ EXACTLY	Derived	
Gravity	GR-IV	$\alpha_1 = -4\lambda_{\text{local}}$; commitment FORCED sparse $\Rightarrow \lambda_{\text{local}} < 1$ (≥ 70 orders safe)	Derived	
Gravity	MetricFromGraph	Reach law $p=1/(d-1) \Rightarrow g \sim 1/b$ in EXACTLY 3D	Derived	
Q-compute	054	UR-1.1/1.2/1.3 three unresolvedness statements ($M_{\text{eff}} < M_{\text{cap}}$; $\Gamma_{\text{cross}} > \Gamma_{\text{decoupling}}$; $\Gamma_{\text{commit}} < \Gamma_{\text{indiv}}^{-1}$)	Derived	
Q-compute	054/056	Existence of Γ_{commit} , $\Gamma_{\text{indiv}}^{-1} \sim c/L_{\text{ED}}$; $M_{\text{cap}} = (\Gamma_{\text{indiv}}/\Gamma_{\text{commit}})^{1/\gamma}$ as the UR-1.1 threshold	Derived	
Q-compute	058b	$\alpha_{\text{topological}} = 0$ (per-locus): the 058 §3.3 saturation never triggers for topological codes	Derived	
Q-compute	058b	Syndrome records are budget-free (P11 commitments)	Derived	
QFT	016	Continuum gauge-covariant derivative $D_\mu = d_\mu + iA_\mu$	Derived	
QFT	017	Lorentz covariance of the continuum free-scalar equation	Derived	
QFT	019	Continuum YM equations $D_\mu F = J$ (Euler-Lagrange)	Derived	
QFT	021	Spectral-gap FORM $\Delta \sim 1/L_*$ (Yang-Mills mass gap)	Derived	
QFT	022	Substrate quotient = physical-state space; BRST = coarse-grained quotient	Derived	
QFT	024	Lindblad/GKSL master-equation form	Derived	
QFT	MS-II	Gauge group $U(N) = SU(N) \times U(1)$ from channel multiplicity	Derived	
QM-kinematics	001	Complex polar carrier $P_K = \sqrt{b_K} e^{i\pi_K}$	Derived	
QM-kinematics	009	Berry phase γ_C = holonomy of the coarse-grained P05 connection	Derived	
QM-kinematics	010	Aharonov-Bohm phase from T17 bundle circulation	Derived	
QM-kinematics	011	Bloch form $\psi_k = e^{i(kx)} u_k(x)$ from discrete translation symmetry	Derived	
QM-kinematics	012.7	Galilean adjacency/propagation bandwidth-asymmetry	Derived	
Relativistic-QM	103	$Cl(3,1)$ frame uniqueness up to similarity	Derived	
Relativistic-QM	104	Anyon prohibition in 3+1D	Derived	
Substrate-evaluation	A1 (CommonCauseNotChannel)	Controlled channel capacity between distinct regions is EXACTLY zero; boundary = common-cause not channel	Derived	
Substrate-evaluation	B4 (ChargeAsTopology)	Winding w in Z conserved + irreversibility-protected; orientation-blindness realizes the skeleton, withholds the field	Derived	
Substrate-evaluation	CleanSubstrateVector	The parity-clean substrate is VECTOR for every channel-count ($\det H(k)$ even $\Rightarrow$ winding=0 for all N); parity violation NECESSARILY spontaneous	Derived	
Substrate-evaluation	QuantumDarwinism	Committed records cost no V5 budget (accounting theorem) $\Rightarrow$ classical objectivity unbounded	Derived	
Constants-inherited	Constants-paper §1	Constants ledger: $c/\hbar/L_P$ present at first events (dimensionful, $\sim$ units); $a_0/\Lambda_{\text{H0}}/\text{masses}$ inherited late (dimensionless, the personality)	Grounded	
Constants-inherited	Constants-paper §3	$G = c^3/L_P^2/\hbar$; G absent from the SM; $[G_d] = m^d$ (units know the dimension count)	Grounded	
Constants-inherited	Constants-paper §4	The dimension-signature in $G(L_P^{d-1})$ IS the holographic area-law / reach-law road — NOT a new independent road to 3D	Grounded	
Cosmology	Cos-01	Inflation = post-SCBU substrate SATURATION regime; $a(t) = a_0 e^{Ht}$	Grounded	
Cosmology	Cos-05	Dark energy = late-universe SATURATION regime; $w = -1$ (leading order)	Grounded	
Cosmology	Cos-05	Λ -smallness (120-OOM) reduces to Route A + Friedmann; naive $\rho_{\text{vac}} \sim M_P^4$ STRUCTURALLY ABSENT substrate-side (no infinite mode-tower)	Grounded	
Cosmology	DF2/DF4	Low-accel galaxies CONSISTENT WITH (not counterexamples to) derived a_0 , via commitment-density saturation suppression	Grounded	
Cosmology	SCBU	a_0 & $x_{\text{canonical}}$ = two projection MECHANISMS of one boundary $R_H = c/H_0$	Grounded	

Domain	Paper	Claim	Tier	Falsifier / kill-condition
Dark-sector	DarkSector	Committed NEUTRAL RELIC candidate for clusters/CMB	Grounded	
Dark-sector	RelicLagrangian	Field content $\Phi = \sqrt{\rho/m_R} e^{i\theta} =$ coarse-grained neutral-channel participation measure	Grounded	
Dark-sector	RelicLagrangian	Khronometric kinetic form L_{kin} (time-along- u + spatial c_{perp})	Grounded	
Dynamics	Dyn-01	Saddle-Hessian-signature dynamics = the foundational generator; 3 regime classes (S1 collapse / S3 inflation-DE / S2 horizon-radiation)	Grounded	
Dynamics	Dyn-02	Horizon-motion law: $RDE(w=1/3)$, $MDE(w=0)$, $LDE(w=-1)$ via substrate Noether stress-energy + DCGT + Friedmann	Grounded	
Dynamics	Dyn-03	Radiation law: EM Larmor + GW quadrupole via Noether flux + DCGT	Grounded	
Dynamics	Dyn-04	Gravitational collapse = monotonic Hessian-signature evolution toward all-compression; horizon = closed signature-class boundary	Grounded	
Dynamics	Dyn-05	Compact-binary inspiral = two-saddle coupled evolution + quadrupole radiation reaction; Peters timescale	Grounded	
Dynamics	GW-00	GW = propagating saddle-Hessian-signature reconfiguration (V1 retarded); energy = local ED-gradient load (no pseudotensor)	Grounded	
Dynamics	GW-00	Quadrupole source-amplitude FORM $P_{\text{GW}} \sim \langle Q\text{-triple-dot}^2 \rangle$	Grounded	
Dynamics	GW-01	BH ringdown = decay of a near-horizon mixed-signature saddle; mode quantization from Hessian eigenstructure	Grounded	
Dynamics	GW-02	Stochastic background $\Omega_{\text{GW}}(f)$ = cosmologically-integrated NoetherFlux over saddle transitions	Grounded	
Entanglement	063	E-2 bilocal cross-chain term Δ_{KL} exists	Grounded	
Gravity	032	Six ED-10 weak-field prerequisites fulfilled at form level	Grounded	
Gravity	033	Scalar-tensor acoustic-metric field eq (covariantization)	Grounded	
Gravity	038.5	$\Lambda = V1$ cosmological-scale integral over R_H ; Λ CONSTANT, $w=-1$ forced, = frozen saturation-floor	Grounded	
Gravity	KM-I	The khronon's deep field IS the MOND sector (unifies the two gravity programs)	Grounded	
Gravity	KM-I	Lensing tracks the MOND potential, no added vector; no new ghost; PPN-safe	Grounded	
Gravity	KM-II	Orthogonality theorem: $\theta=0$ in statics $\Rightarrow$ the regulator family is invisible to every tested result	Grounded	
Gravity	KM-II	Clustering dial = the soft spot (one dial sets degeneracy-cure + Cerenkov + clusters/CMB)	Grounded	
Gravity	KM-II	One-Lambda thesis: $W0 = -24\pi^2\Omega\Lambda \sim -162$	Grounded	
Q-compute	053	$M_{\text{cap}} = \min(N_{\text{bw}}, N_{\text{V5}}, N_{\text{commit}})$ is one substrate object; min-structure forced by simultaneity	Grounded	
Q-compute	054	The three UR-1 rate quantities have distinct PRIMARY DRIVERS (basis of A/B/C distinctness)	Grounded	
Q-compute	055	A/B/C three-class EXHAUSTIVENESS — every architecture saturates one of the three constituents	Grounded	
Q-compute	057	Class-B exponential gap-suppression $\Gamma_{\text{fail}} \sim \Gamma_0 e^{-\xi/\xi_0}$	Grounded	
Q-compute	058	Class-C plateau $\Gamma_{\text{fail}} \rightarrow \Gamma_{\text{plateau}} > 0$ at redundancy saturation	Grounded	
Q-compute	059	Meta-architectures (EC/DD/reservoir/hybrids) = compositions that shift which constituent binds -- NO fourth class	Grounded	
Q-compute	060	M_{crit} unification: the matter-wave Q-C boundary and the qubit walls are one substrate threshold	Grounded	
QFT	013	V1 QFT-arc propagator (Kallen-Lehmann retarded form)	Grounded	
QFT	014	$N1+GR1$: V1 finite-width + 2-point function on a curved acoustic background	Grounded	
QFT	017	Free scalar Klein-Gordon $(\Box+m^2)\phi=0$ as the DCGT limit of V1	Grounded	
QFT	018	Non-abelian YM potential, field strength, YM equation	Grounded	
QFT	MS-II	Substrate is a $U(N)$ lattice gauge theory; escapes Nielsen-Ninomiya	Grounded	
QFT	MS-II	Spin-1/2 from relationality (tethered node carries $SU(2) \rightarrow SO(3)$ double cover)	Grounded	
QFT	MS-II	Parity violation is non-abelian: NO parity-violating abelian force	Grounded	

Domain	Paper	Claim	Tier	Falsifier / kill-condition
QFT	MS-II	Chirality γ^5 = the arrow's 'screw'	Grounded	
QFT	MS-II	$\{1,2,3\} \Leftrightarrow$ internal amplitude dimension $d=3$ (Welch-bound); no stable $SU(N>=4)$	Grounded	
QFT	MS-II	Three spatial dimensions because the arrow must hold committed order	Grounded	
QM-kinematics	003	Born rule $f_K = c_K ^2$ as the participation-frequency limit	Grounded	
QM-kinematics	003.5	$P_K = \sqrt{b_K} \cdot e^{i \phi_K}$ amplitude carrier, complex-valued forced	Grounded	
QM-kinematics	004.5	Sesquilinear inner-product form $\langle P, Q \rangle = \sum P_K^* Q_K$ (discrete)	Grounded	
QM-kinematics	004.6	Continuum inner-product $\int \Psi^* \Phi$ dm on $L^2(M, dm)$	Grounded	
QM-kinematics	005	Projective measurement (projector P_i , collapse, Born prob)	Grounded	
QM-kinematics	005.5	Double-slit interference $\rho = \sum P_K ^2$; which-path = V5 injection	Grounded	
QM-kinematics	006	Unitary evolution $\rightarrow$ Schrodinger $i \hbar \partial_t \Psi = H \Psi$	Grounded	
QM-kinematics	006.5	Schrodinger equation via Stone's theorem	Grounded	
QM-kinematics	006.6	Kinetic operator $T = p^2/2m$, the factor-2	Grounded	
QM-kinematics	006.7	Schrodinger emergence in the thin-participation limit	Grounded	
QM-kinematics	007	Hilbert space = Cauchy completion of the motif algebra	Grounded	
QM-kinematics	008	Phase structure $S^1/U(1)$ on rule-types	Grounded	
QM-kinematics	008.5	Phase-independence of bandwidth = $U(1)$ gauge invariance	Grounded	
QM-kinematics	012.5	Momentum operator $p = -i \hbar \text{grad}$ = translation generator	Grounded	
QM-kinematics	012.6	Heisenberg $Dx \cdot Dp \geq \hbar/2$	Grounded	
Relativistic-QM	102	Spin-statistics $\eta = (-1)^{2s}$	Grounded	
Relativistic-QM	106 (T4)	Dirac equation $(i \gamma^\mu \partial_\mu - mc/\hbar) \Psi = 0$	Grounded	
Relativistic-QM	107	Klein-Gordon equation $(\Box + m^2 c^2/\hbar^2) \Psi = 0$	Grounded	
Relativistic-QM	108	Minimal-coupling KG + conserved 4-current	Grounded	
Relativistic-QM	109	Lorentz-representation taxonomy	Grounded	
Relativistic-QM	111	Primitive-level UV finiteness	Grounded	
Relativistic-QM	112 (T10)	Lightlike worldlines for $\sigma=0$ rule-types	Grounded	
Relativistic-QM	113	Mass structural form (Arc-M H1)	Grounded	
Relativistic-QM	114 (GRH-D1)	Existence of ≥ 1 massless Case-P gauge rule-type	Grounded	
Relativistic-QM	115	Vacuum + particle dual Fock structure ($\sigma=0$)	Grounded	
Relativistic-QM	116	MR-R / MR-P conditional massless slot	Grounded	
State-reduction	ChannelGranularity	ED's channel/pointer basis is COARSE ($> I_{ED}$), so bound charges are single-channel	Grounded	
State-reduction	CollapseRate	$\tau \sim \hbar/E_G$ SCALING from ED's primitives (two branch commitment-clocks $N \sim \sqrt{b}$) the single arrow cannot hold $\rightarrow$ einselection individuates)	Grounded	
State-reduction	CollapseRate	Two refinements over the rivals: a DEFINITE reduction (not Diosi's mere decoherence) + a NATIVE regulator for the point-mass divergence	Grounded	
State-reduction	CollapseRate	NOT excluded by the 2020 spontaneous-radiation bound (Lindblad jump operators = channel projectors $J = g \cdot P_i$, FORCED from P11; committed states = exact fixed points $D[P_i]=0$; no continuous diffusion)	Grounded	
State-reduction	OpenPiece-i	MOND enhancement of E_G (mesoscopic masses sub- a_0 in self-gravity $\rightarrow$ naive $\sim 5\text{-}50\times$) is CLOSED by ED's own External Field Effect (Paper_029 horizon dipole sourced by total accel; Earth Newtonizes to $\sim 3.5e-6 \rightarrow$ ED $\sim$ Penrose-Diosi in the lab)	Grounded	
State-reduction	Positioning §3.2/§6	Locality WITHOUT superdeterminism (local commitment + A1 capacity-zero + stochastic-at-commitment; no hidden variable)	Grounded	
Substrate-evaluation	A2 (GrownNotInstalled)	A grown decoupling boundary reproduces exact channel-zero	Grounded	

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Substrate-evaluation	B4/Maxwell	Integral Gauss law circulation= $2\pi w$ (loop-independent); the Maxwell action is LATENT in the coherence deficit $(\sin^2(d\phi/2) \sim 1/4(\text{grad } \phi)^2)$	Grounded	
Substrate-evaluation	CleanSubstrateVector	$\gamma^5 = \text{arrow} \times \text{spontaneous spatial orientation}$; the vector clean substrate is automatically anomaly-free	Grounded	
Substrate-evaluation	GaugeStructure	Gauge group $U(N)=SU(N)\times U(1)/Z_N$ form; F^2 action; mass-gap MECHANISM [A,A] lifts flat direction iff non-abelian; single $U(1)_Y$	Grounded	
Substrate-evaluation	Gleason keystone	Channel orthogonality REDUCES to channel-distinctness given the representation; einselection primitive; Born non-contextuality; C selected	Grounded	
Substrate-evaluation	V5AttractiveSign/PhaseCoherence	V5's coupling IS the coherence content of the cross-chain superposition ($\sqrt{b_A b_B} \cos(d\phi - \text{int } A)$); MOND cross-term constructive (P12-Coh rewards alignment)	Grounded	
Substrate-evaluation	V5UnifiedBudget/QuantumDarwinism	$\rho_{\text{loc}}=1$: monogamy budget = Class-C plateau budget (one W_{max} spent two ways); per-locus live cap $N_{\text{live}} \leq W_{\text{max}}/w_{\text{bar}}$	Grounded	
Black-hole	HorizonTiling	Horizon tiles ~ 1 bit/Planck cell (3 counts: holo=1, frozen 0.78, edge 0.88)	Measured	
Gravity	GR-III	Dynamical rule $b \cdot \dot{b} = D \text{grad}^2 b - \kappa \rho$: steady state $\text{grad}^2 b \sim \rho$ (0.999); $r_s \sim M$; emergent frozen $b > 0$ horizon; area law $S \sim A$	Measured	
Gravity	GR-III	Hawking scaling $\kappa = 1/(2 r_s) \sim 1/r_h$	Measured	
Gravity	QuadraticStrain	Phase-coherence constructive sign supplied (P12-Coh rewards alignment)	Measured	
QFT	MS-II	Linking held in 3D, erasable in 4D	Measured	
QFT	MS-II	Retarded ('arrow') hopping produces net chirality signatures	Measured	
Substrate-evaluation	AreaLaw/V5UnifiedBudget	Straddling cross-chain (V5) edge count scales as AREA ($r^{2.02}$), reach-independent; holographic entropy = the edge count ($\sim 0.88/\text{Planck cell}$)	Measured	
Substrate-evaluation	KineticLatticeGas/CanonicalPDE	Certified continuum = kinetic lattice-gas (BALLISTIC), NOT a diffusion PDE; and does NOT generate the UDM mobility law ($\beta \sim 0$ not 2)	Measured	
Substrate-evaluation	KnotsNotCrystals/RelationalTick	No spontaneous long-range correlation-order in any sector; persistence is rhythmic (V5 binds local proper time, preserving time dilation)	Measured	
Substrate-evaluation	MassWithoutMass	Binding confines massless fronts into a sub-luminal inertial composite (mass without mass); $\text{mass}(\text{binding}) \neq \text{time-dilation}(\text{commit-rate})$	Measured	
Substrate-evaluation	Maxwell	The coherence-weighted continuum limit IS Coulomb, but the DETERMINATE substrate does NOT cast the smooth field (emergent shadow)	Measured	
Substrate-evaluation	PhaseCoherence/V5AttractiveSign	P12 coherence rewards alignment but stays FINITE-REACH (no crystal) 2D+3D; V5's attractive local-lock from P12's positive sign (flip destroys it)	Measured	

The Rest of the Register (168 rows — inherited values, syntheses, walls, and honest open problems)

Selected/Inherited (41)

A value taken from measurement, or a principled selection among a theorem's allowed options — not substrate-derived.

Domain	Paper	Claim	Tier	Falsifier / kill-condition
Black-hole	043	Bekenstein-Hawking 1/4 coefficient, COMBINATORIAL route	Selected/Inherited	
Black-hole	044_5	Greybody factors Γ_{ls}	Selected/Inherited	
Black-hole	047_5	Universal $T = \kappa/2\pi$ for all 4 horizon types	Selected/Inherited	
Black-hole	050	$t_{\text{Page}} \sim 0.54 \tau_{\text{BH}}$; c_1 , w , c^* values	Selected/Inherited	
Cosmology	Baryogenesis	First-arrival chirality χ^* (which template = matter)	Selected/Inherited	
Cosmology	Cos-01/05/06	Uniform- Ψ realizability (the saturation IC)	Selected/Inherited	
Cosmology	Cos-03/04	$\Omega_{\text{c}} h^2 \sim 0.12$ (CMB cold-dark-matter component, $\sim 5x$ baryons)	Selected/Inherited	
Cosmology	all Cos	H_0 value	Selected/Inherited	
Dark-sector	DarkSector	Bullet Cluster offset (relic sails through, collisionless)	Selected/Inherited	
Dark-sector	RelicLagrangian	Relic mass m_{R}	Selected/Inherited	
Dark-sector	RelicLagrangian	Relic abundance $\Omega_{\text{c}} \sim 0.12$	Selected/Inherited	
Dynamics	Dyn-03	Larmor coefficient $1/(6 \pi \epsilon_0 c^3)$; quadrupole coefficient $1/(5c^5)$	Selected/Inherited	
Dynamics	GW-00	Two transverse polarizations (h_+ , h_x); no scalar/vector/longitudinal	Selected/Inherited	
Dynamics	GW-00/GW-01	GW speed $c_{\text{T}} = c$	Selected/Inherited	
Dynamics	GW-01	QNM frequency spectrum $\omega_{\text{lmn}}(M, J) \sim c^3/GM + \text{Regge-Wheeler/Teukolsky machinery}$	Selected/Inherited	
Dynamics	all	$h \sim 1/r$ amplitude falloff; $L_{\text{saddle}}/\tau_{\text{V5}}/\Theta_{\text{ED}}/H_0$ (Route A4); $r_{\text{horizon}}=2GM/c^2$, $r_{\text{ISCO}}=6GM/c^2$	Selected/Inherited	
Entanglement	064	The Schmidt decomposition form itself	Selected/Inherited	
Entanglement	068	$k_{\text{B}} / \log$ -base normalization; concrete entropy values	Selected/Inherited	
Entanglement	069	Tsirelson bound $2\sqrt{2}$	Selected/Inherited	
Entanglement	069	Bell < Tsirelson < NS nesting; PR-box exclusion	Selected/Inherited	
Foundations	087/089	$L_{\text{ED}} = L_{\text{P}}$ (Planck length)	Selected/Inherited	
Foundations	090/089	τ_{V5} memory times (τ_{soft} =measured Maxwell time; $\tau_{\text{BH}}=L_{\text{P}}/c$; τ_{ent} system-specific); V_1/V_5 envelope shapes	Selected/Inherited	
Foundations	096/097	$\xi_{\text{canonical}} = 1.7575 \text{ lu}$ (cross-scale operating point); the 0.6 three-regime-RG transition exponent	Selected/Inherited	
Gravity	027	$G = c^3 L_{\text{P}}^2 / \hbar$	Selected/Inherited	
Gravity	029	$a_0 = cH_0/(2\pi)$	Selected/Inherited	
Gravity	038.5	Lambda magnitude $\rho_{\text{Lambda}} \sim 1.1e-52 \text{ m}^{-2}$	Selected/Inherited	
Q-compute	053-058	Substrate coefficients Γ_{ma0} , γ , α , β , ξ_0 , Δ_{ta0} , $B_{\text{cross,max}}$, $R_{\text{C,sat}}$, Γ_{plateau}	Selected/Inherited	
Q-compute	056	140-250 kDa Class-A wall mass (central $\sim 180 \text{ kDa}$)	Selected/Inherited	
Q-compute	057/058	Specific codes (surface, Majorana, Shor, Steane, concatenated); rigidity/redundancy params ξ , R_{C}	Selected/Inherited	
QFT	015-019	Empirical gauge groups $SU(3) \times SU(2) \times U(1)$, couplings, masses, mixings	Selected/Inherited	
QFT	015.5	Integer-Chern quantization C_{n} in $\mathbb{Z}$ of photonic Hall drift	Selected/Inherited	
QFT	019/021/024	g_{YM} , mass-gap coefficient, Lindblad L_{k} & rates	Selected/Inherited	
QM-kinematics	004.5	Tsirelson bound $2\sqrt{2}$	Selected/Inherited	
QM-kinematics	012.5/012.6/006.5/006.6	$\hbar$ throughout the operator sub-arc	Selected/Inherited	
QM-kinematics	Gleason-Recon (sibling)	Complex field C (from the Soler menu $\{R, C, H\}$)	Selected/Inherited	

Domain	Paper	Claim	Tier	Falsifier / kill-condition
Relativistic-QM	110 (T8)	Canonical (anti-)commutation $[x,p]=i\hbar$	Selected/Inherited	
Relativistic-QM	112 ($\hbar$)	$\hbar$ origin = participation quantum per chain-step	Selected/Inherited	
Substrate-evaluation	CleanSubstrateVector	The casting (which force is chiral) = SU(2) fundamental pseudoreal vs SU(N>=3) complex	Selected/Inherited	
Substrate-evaluation	GaugeStructure/V5UnifiedBudget	Couplings g, scale a, charge values, EWSB; absolute V5 budget scale $W_{\max}$ (=Gamma_plateau, t_Page, CKW coeff)	Selected/Inherited	
Substrate-evaluation	Gleason keystone	Number field C selected from Soler's {R,C,H}	Selected/Inherited	
Substrate-evaluation	ParityWall	Matter/antimatter reference (C) natively selected (first-arrival P09-phase)	Selected/Inherited	

Standing ⁽⁷⁾

A settled result from another ED arc, cited here and built on — not re-derived in this folder.

Domain	Paper	Claim	Tier	Falsifier / kill-condition
Dark-sector	(KM-I/027-034)	Galactic dynamics from MOND/horizon, no dark-matter particle	Standing	
Dark-sector	031	Baryonic Tully-Fisher slope-4 + near-zero scatter	Standing	
Dark-sector	DarkSector	Cluster missing-mass shortfall filled by the relic	Standing	
Dark-sector	QuadraticStrain/030	MOND = the off-diagonal HORIZON-INTERFERENCE cross-term; $\sqrt{a_N a_0}$ forced	Standing	
State-reduction	Positioning (Report §4)	Arrow selects the pointer basis (einselection PRIMITIVE); repeatability + Born non-contextuality follow	Standing	
State-reduction	Positioning (Report §6)	Unitarity emergent, not fundamental (arrow forbids erasure; between-commitments = reversible QM)	Standing	
State-reduction	Positioning §7	Geometry emergent $g \sim 1/b$ (both matter and geometry emergent)	Standing	

Reference ⁽⁶⁾

A canonical DEFINITION or catalogue (a kernel, the verdict grammar, an audit) — scaffolding, not a claim to be proven.

Domain	Paper	Claim	Tier	Falsifier / kill-condition
Foundations	088	Primitive Load-Bearing Audit (per-primitive dependency map + per-primitive falsifiers)	Reference	
Foundations	089	V1 = the single-chain finite-width retarded vacuum kernel $K_{V1} = \theta(t-t')G(\sigma/L_{ED}^2)$	Reference	
Foundations	090	V5 = the cross-chain correlation kernel (finite-memory τ_{V5} , retarded, U(1)-compatible)	Reference	
Foundations	095	Form-FORCED / value-INHERITED methodology; the M1/M2/M3 verdict tiers (+ label classes P/A/I/D/D-via-I/OPEN)	Reference	
Foundations	095	I sub-labels (substrate-parameter-INHERITED, IC-INHERITED) + mixed-tier / component-composition rules	Reference	
QFT	013	V1 finite-width retarded vacuum kernel (QFT-arc presentation)	Reference	

Inherited / Consilience ⁽⁵⁾

Reproduces a known result/sequence framed THROUGH the substrate — supporting consilience, NOT a novel prediction.

Domain	Paper	Claim	Tier	Falsifier / kill-condition
Cosmology	Cos-02	BBN = freeze-out competition Γ_{nuc} vs substrate-H(t); $Y_p \sim 0.247$	Inherited / Consilience	
Cosmology	Cos-03	CMB acoustic peaks = angular projection of baryon-photon oscillator; $L_n \sim n\pi D_A/r_s$	Inherited / Consilience	
Cosmology	Cos-04	Linear structure = growth of primordial perturbations $P(k,a)=D^2 T^2 P_{\text{init}}$	Inherited / Consilience	
Cosmology	Cos-06	Inflationary fluctuation spectrum = sub-leading $O(\epsilon)$ perturbations on Cos-01 saturation	Inherited / Consilience	
Soft-matter	P04TransportBudget_Resistivity	Resistivity saturation = Mott-Ioffe-Regel limit re-grounded in bounded P04 participation: finite per-cell bandwidth floors the mean free path at the effective grain a_{eff}	Inherited / Consilience	resistivity keeps rising with no saturation once loss controlled (MIR already observed -> consilience pass, not a new bet)

Native / V5-conditional (1)

Mechanism is native but rests on V5, a faithful structural addition not shown primitive-forced.

Domain	Paper	Claim	Tier	Falsifier / kill-condition
Dark-sector	RelicLagrangian	Relic is MASSIVE by V5 binding, cold-capable	Native / V5-conditional	

Identification (3)

A structural mapping onto an existing EXTERNAL framework (e.g. Penrose CCC, Jacobson's equation of state).

Domain	Paper	Claim	Tier	Falsifier / kill-condition
Cosmology	CCC	CCC's six structural elements <-> ED substrate content (X<->SCBU; twistor<->V5 memory; next-aeon<->P12 re-entry; etc.)	Identification	
Cosmology	CCC	Contrast-zero state unique ==> CCC crossover GLUING is an identity, not a postulated match	Identification	
Cosmology	CCC	Planck-mass remnants dissolve at X (Scenario-C cutoff = two-chain V5 handshake; boundary emptying removes the partner)	Identification	

Synthesis (31)

An arc consolidation / overview / organizing lens; composes standing results, adds no new derivation.

Domain	Paper	Claim	Tier	Falsifier / kill-condition
Black-hole	047_5	Cross-class horizon identification (M3)	Synthesis	
Black-hole	052	Paradox synthesis (integrates 039+047+050+041)	Synthesis	
Dynamics	Dyn-01	Cos-01 (inflation) + Cos-05 (dark energy) = the single all-expansion (S3) saddle regime at different epochs	Synthesis	
Dynamics	Dyn-05/Dyn-04/GW-01/GW-02	Full inspiral->plunge->collapse->ringdown waveform closed at corpus level; BH horizon + cosmic decoupling = one substrate-parameter surface	Synthesis	
Entanglement	070	Bipartite architecture as one phenomenon	Synthesis	
Foundations	091/092	Memory-kernel cascade V1(UV)->V5(memory) under DCGT; kernel hierarchy (three-index; V1=(L_ED,1,1), V5=(L_V5,inf,2))	Synthesis	
Foundations	096/097	Cross-scale invariance (S1 kernel-content invariant); three-regime RG (UV/transition/IR)	Synthesis	
Foundations	098	ED-QFT unified overview (5 sectors); Phase-1 QM at M2, DCGT M3, NS/MHD/YM/gravity M2-class	Synthesis	
Foundations	099	NS / NS-MHD synthesis — NS-smoothness at Intermediate Path C / structural-positive (register identical to YM Clay-relevance); F3 = constructive blowup refutes	Synthesis	
Foundations	100	Five-sector program overview (one substrate / five sectors / one methodology)	Synthesis	
Gravity	OneField	'One field, one scale, five roles' (arrow, Einstein sector, MOND, clustering dial, Lambda-scale)	Synthesis	
Gravity	QuadraticStrain	MOND = interference cross-term: Newton=diagonal, MOND=off-diagonal of sum P ^2	Synthesis	
Other-theories-advanced	GravityAsHorizonEquipartition	Grounds Padmanabhan's holographic equipartition (= ED's Theorem T19): ED supplies the count N_sur=A/L_P^2 (A1 severance + the measured edge-count area law)	Synthesis	
Other-theories-advanced	InducedGravityArena_Sakharov	Grounds Sakharov induced gravity: ED supplies the presupposed manifold + cutoff (arena from a real substrate; grain=cutoff) + a substrate-native rigidity of gravitational strength	Synthesis	
Other-theories-advanced	KhronometricEquationOfState_Jacobson	The local-Rindler delta-Q=T-dS derivation for a khronometric theory: on ED's luminal family Wald entropy is pure area -> clean equilibrium -> FULL Einstein (aether among sources); shear stays exactly GR (c13=0)	Synthesis	
Other-theories-advanced	MoreThanAnalogy_AnalogGravity	Grounds analog gravity's 'is there a real medium?': ED's emergent metric IS an acoustic metric (GR-I) with a REAL medium (the substrate) + the DYNAMICAL aether (the khronon) analog models lack	Synthesis	
Other-theories-advanced	WhyTheCountIsArea_Susskind	Grounds the 't Hooft/Susskind holographic 'why area, not volume?': A1 severs participation wires strictly at a surface + the short-range kernel makes the straddling-edge count grow like AREA (measured r^2.02)	Synthesis	
Q-compute	061	Q-Compute arc consolidation (053-060); 'decoherence-centric -> multiplicity-centric' reframing	Synthesis	
Q-compute	062	Cross-domain echo: the BH horizon Gamma_cross->0 collapse and Q-Compute M_cap saturation share one V5 mechanism	Synthesis	

Domain	Paper	Claim	Tier	Falsifier / kill-condition
QFT	023	Yang-Mills arc closes at 'Clay-relevance / structural-positive level'	Synthesis	
QM-kinematics	011.5	Four QM postulates unified under the participation measure	Synthesis	
State-reduction	Positioning	THE arrow = gravity's time = the pointer-basis selector — one primitive, two hats (collapse variable = TIME not energy)	Synthesis	
State-reduction	Positioning §3.2	ED supplies the well-defined preferred time whose absence drives Penrose's argument (collapse upstream of E_Delta)	Synthesis	
Substrate-evaluation	ArrowSortsTheContinuum	The arrow sorts continuum laws into three classes (structure-making native / erasing at layer-2 / preserving = edge)	Synthesis	
Substrate-evaluation	GravityAsEquationOfState	ED grounds Jacobson's 3 assumed inputs (severance/grain/eta) and lands $G=c^3 L_P^2/\hbar$	Synthesis	
Substrate-evaluation	HowPDECoarseGrainReality	Two hard shadows: YM F^2 derived-given-inputs + gap mechanism-tier (continuum OPEN); NS no-blow-up on 2 postulates; finite-grain singularity triple P04+V5+P11	Synthesis	
Substrate-evaluation	HowTheoryCoarseGrainReality	Each best theory = a blur of one bandwidth field (kept->variables, discarded->symmetries); G/GR at the seam; Born un-blurred; thermo the receipt	Synthesis	
Substrate-evaluation	KnotsNotCrystals	ED has topological (knot/link) order, NOT crystalline	Synthesis	
Substrate-evaluation	MinimalOntology	Shortest naming of ED = 'commitment and participation' over a spatial stage (floor = 3 substrate + 4 arena + 1 commitment)	Synthesis	
Substrate-evaluation	V5UnifiedBudget	Monogamy(065)+Page(050)+Class-C(058) = projections of one bounded envelope $W_{\max}$; R2 ratios 1:1:0.88	Synthesis	
Soft-matter	UnifiedP04TransportBudget	Unified P04 transport budget: viscosity floor & resistivity ceiling are ONE Adjacency-band wall (both = mean free path floored at a_{eff}); ED unifies MIR + Planckian dissipation + KSS viscosity bound as one substrate event	Synthesis	F-COONSET: the two onset at different occupancies; F-BAND: charge & momentum use distinct channels not one shared Adjacency band

Asserted (16)

An A->position / A->assertion: stated in the paper's own audit as a position, NOT proven (the weakest claim tier).

Domain	Paper	Claim	Tier	Falsifier / kill-condition
Black-hole	041	'Remnants NOT a dark-matter candidate'	Asserted	
Black-hole	042	'Classical singularity is a coarse-graining artifact'	Asserted	
Black-hole	044_5	'Leading-order match is identification, not derivation'	Asserted	
Black-hole	047_5	'Four-horizon structural identification'	Asserted	
Black-hole	051	'Paradox not generated' + 'information not lost'	Asserted	
Constants-inherited	Constants-paper §2	$\hbar$ -as-grain-of-memory; becoming/memory/extent ($c/\hbar/L_P$); G-as-first-shadow	Asserted	
Dynamics	GW-00/GW-01/Dyn-04	No graviton, no metric perturbation, no Hilbert-mode, no gauge freedom, no pseudotensor	Asserted	
Entanglement	063	E-2 non-factorizability is IRREDUCIBLE	Asserted	
Entanglement	071	ER=EPR-class echo	Asserted	
Entanglement	072	Entanglement = unresolved individuation regime (not action-at-a-distance)	Asserted	
Other-theories-advanced	(folder finding)	Gravity's asymmetry: gravity is the unique arrow-throwing, non-obviously-thermodynamic theory cleanly recoverable as an equation of state (no famous EM/Schrodinger version)	Asserted	
QFT	022	Gribov ambiguity is a coarse-graining artifact (not constructively resolved)	Asserted	
Relativistic-QM	106/107/108/111/113	Arc-R stage placements + M-verdict classification rows	Asserted	
Relativistic-QM	111	'UV divergences = coarse-graining artifacts of $\omega_c \rightarrow \infty$ '; renorm = substrate-cutoff-absorption	Asserted	
Substrate-evaluation	MinimalOntology	Commitment (P11) is the SOLE process primitive (ED has a spatial arena but no separate temporal one)	Asserted	
Substrate-evaluation	ProxyConversionDoctrine	Proxy Conversion Doctrine D1-D4 (one rule set replacing five local patches)	Asserted	

Wall (4)

A proven or honestly-checked LIMIT — the order or result ED does NOT reach (a load-bearing 'no').

Domain	Paper	Claim	Tier	Falsifier / kill-condition
Substrate-evaluation	ArrowSortsTheContinuum	Solitons: the bare substrate does not build a reversible balance	Wall	
Substrate-evaluation	GaugeStructure	Multiplicities {1,2,3} and anomalies NOT selected	Wall	
Substrate-evaluation	ParityWall	The weak force's chirality (P) is NOT derived — found absent at every level checked	Wall	
Substrate-evaluation	Primes (FiniteMemoryCeiling)	Finite memory saturates the prime template but PROVABLY NOT the escape (parity layer)	Wall	

Open (51)

A declared OPEN / unclosed item — a named gap or frontier.

Domain	Paper	Claim	Tier	Falsifier / kill-condition
Black-hole	042/046	Substrate BH interior for $r < l_P$	Open	
Black-hole	043	Substrate derivation of $\log g$	Open	
Black-hole	045	Superradiance	Open	
Black-hole	050/051	Full unitarity (constructive, not postulated)	Open	
Black-hole	052_5	Substrate-anchored τ_{V5} floor (merger-lag)	Open	22-Ways P2 (merger-lag): PROVISIONAL, existence-only; NOT refutation-grade (needs substrate τ_{V5} floor, OPEN)
Black-hole	BH_Thermal2Pi	Arrow-native (continuation-free) 2π	Open	
Black-hole	HorizonTiling/046	Curvature emergence underneath the geometric counts	Open	
Cosmology	CCC	Twistor-analog substrate integral; Big-Ring/Grand-Arc scale; cyclic vs single-aeon termination	Open	
Cosmology	Cos-01/05	Route A (substrate-derived $l_{V5}(H_0) \rightarrow$ quantitative Λ , $N \sim 60$, H)	Open	
Cosmology	Cos-02/03/04/06	Non-load-bearing inherited puzzles: 7Li, Hubble tension, S8, ~ 60 e-folds / termination	Open	
Cosmology	Cos-03/04	CMB/clusters decoupled dark component has NO native ED mechanism	Open	
Cosmology	DF2/DF4	Suppression function (ED analogue of MOND's ν) from primitives	Open	
Cosmology	README	Gaussianity / CMB observed-layer debt	Open	
Cosmology	SCBU	$\chi_{\text{canonical}}$ substrate-derivation from H_0 (analogous to a_0)	Open	
Dark-sector	DarkSector	CMB spectrum computed from ED	Open	F6: a computed CMB spectrum from the relic action that misses the Planck peaks for EVERY mass in the warm window
Dark-sector	DarkSector	Galactic small-scale puzzles (missing-satellite, satellite-plane)	Open	
Dark-sector	DarkSector	DF2/DF4 suppression function	Open	
Dark-sector	RelicLagrangian	Relic mass / free-streaming value	Open	
Dark-sector	RelicLagrangian	Non-perturbative / vortex edge of L_{self}	Open	
Dynamics	Dyn-02	OPEN-HM-Q1 (chain-class criterion rapid- V_1 =RDE vs slow- V_1 =MDE) + OPEN-HM-Q2 (stress-energy form per regime)	Open	
Dynamics	Dyn-03	OPEN-RL-Q1 (source-class criterion) + OPEN-RL-Q2 (flux form per class) = THE single shared gap of the folder	Open	
Entanglement	064	derivability of P-V5-Schmidt-Generic	Open	
Entanglement	068	O-vN-1: derive P-Additivity from P04 (quantitative)	Open	
Entanglement	068	O-vN-2: derive P-Maximality from P11	Open	
Foundations	087/088/092/093	P11 from a deeper layer (T18 $M_2 \rightarrow M_1$); minimality of the 13; higher-order kernels V_n ; P14 status	Open	
Foundations	090	Forward derivation of V5 existence (currently a P10-licensed posit)	Open	

Domain	Paper	Claim	Tier	Falsifier / kill-condition
Foundations	098/100	Substrate-level derivation of $\hbar$, c , G , a_0 ; Phase-3 Einstein-equation emergence (still SPECULATIVE)	Open	
Gravity	038.5	Whether Lambda frozen-floor value is FORCED or stipulated	Open	
Gravity	038.5/KM-II	Ab-initio V1 boundary integral for Lambda (target $W_0 = -24\pi^2 \Omega_L$); Route-A closure	Open	
Gravity	KM-I/KM-II	Clusters + CMB not discharged; khronon-standalone ABUNDANCE GAP	Open	
Gravity	MetricFromGraph	Nonlinear / strong-field field equations; background-free construction	Open	
Other-theories-advanced	(Wallstrom / Entropic Dynamics)	ED grounds the ad-hoc integer single-valuedness of Entropic Dynamics (Caticha) / resolves the Wallstrom objection	Open	
Q-compute	054	Formal mathematical independence of the three UR-1 rate quantities	Open	
Q-compute	054	The power-law form of $\Gamma_{\text{commit}}(M_{\text{eff}})$	Open	
Q-compute	056	Strategy-1 substrate-derived platform-scaling commitments (cross-platform unification)	Open	
Q-compute	057	Substrate-level derivation of P-Rigidity-Gap + P-Boltz-Analogy from primitives	Open	
Q-compute	058b	The repetition-code-floor discriminator (does a residual floor survive burst mitigation at cross-platform-consistent content?)	Open	
QFT	019/020/021/023	YM continuum measure; OS3 derivation; mass-gap survival as $a \rightarrow 0$ (the 3 Clay-hard gaps)	Open	
QFT	021/023	Confinement, asymptotic freedom, glueball spectrum, numerical Delta	Open	
QFT	MS-I/MS-II	Yang-Mills action ($\sim F^2$); electroweak/Higgs sector; EWSB; fermion masses	Open	
QFT	MS-II	{1,2,3} uniqueness; weak-force specific chirality; anomaly cancellation; the linking premise	Open	
QFT	MS-II	Full channel-topology $\rightarrow$ representation spectrum; spinor construction	Open	
QM-kinematics	004	Substrate Gleason: inner-product uniqueness	Open	
QM-kinematics	Gleason-Recon (sibling)	Orthogonality REDUCED; C selected; residual = Soler	Open	
State-reduction	ChannelGranularity	Explicit channel-granularity theorem + exact distinguishability floor	Open	
State-reduction	CollapseRate	Exact collapse coefficient	Open	
State-reduction	CollapseRate/OpenPiece-i	Exact energy functional (from ED's quadratic-strain gravity, not Einstein-Hilbert)	Open	
Substrate-evaluation	B4/Maxwell	Whether DCGT coarse-graining selects the Maxwell-action configuration	Open	
Substrate-evaluation	CleanSubstrateVector	Relativistic substrate $\rightarrow$ Dirac descent	Open	
Substrate-evaluation	Gleason keystone	Soler lattice technical rigor (exchange-property geometry, inf-dim)	Open	
Substrate-evaluation	V5AttractiveSign/RelationalTick	Forward derivation of V5 from the primitives; the temporal envelope shape F_{V5}	Open	

Superseded (3)

Retracted / withdrawn / superseded by a later corpus result (kept for provenance).

Domain	Paper	Claim	Tier	Falsifier / kill-condition
Cosmology	Baryogenesis	P-BinaryAdmission	Superseded	
Dark-sector	RelicLagrangian	L_{self} 'open superfluid self-term' + the khronon-mode one-substance probe	Superseded	
Entanglement	067	von Neumann E-6 (duplicate)	Superseded	

Tier Key (all 19 tiers)

Rank	Tier	Definition	Kind	Count
1	Standing	A settled result from another ED arc, cited here and built on — not re-derived in this folder.	Claim	7
2	Derived	Forced/proven from the 13 primitives (+ standard math) with NO paper-specific postulate — the strongest claim tier (theorems; M1).	Claim	55
3	D-via-I / Form-forced	The FORM is derived by composing inherited/standard pieces (derived-via-inherited); the form is forced, the numbers inherited.	Claim	13
4	Grounded	Structural / conditional / account-tier: form-forced GIVEN a stated input or a declared paper-specific postulate (M2/M3).	Claim	85
5	Reference	A canonical DEFINITION or catalogue (a kernel, the verdict grammar, an audit) — scaffolding, not a claim to be proven.	Input	6
6	Measured	A simulation / build-and-run / probe result on the certified substrate (or a faithful model of it).	Claim	12
7	Inherited / Consilience	Reproduces a known result/sequence framed THROUGH the substrate — supporting consilience, NOT a novel prediction.	Claim	4
8	Native / V5-conditional	Mechanism is native but rests on V5, a faithful structural addition not shown primitive-forced.	Claim	1
9	Selected/Inherited	A value taken from measurement, or a principled selection among a theorem's allowed options — not substrate-derived.	Input	41
10	Constant	A fundamental constant, by when it enters: event-scale (units, present at the first events) / derived-combination / inherited-late.	Input	10
11	Primitive	One of the 13 postulated axioms (P01-P13) — the corpus's single 'conditional on'; postulated, not derived, not claimed minimal.	Input	13
12	Postulated	Rests on a named paper-specific postulate (a declared commitment BEYOND the 13 primitives).	Input	24
13	Asserted	An A->position / A->assertion: stated in the paper's own audit as a position, NOT proven (the weakest claim tier).	Claim	16
14	Prediction	A falsifiable, experiment-facing bet — carries a kill-condition (see the Falsifier / Kill-Cond column).	Claim	23
15	Identification	A structural mapping onto an existing EXTERNAL framework (e.g. Penrose CCC, Jacobson's equation of state).	Claim	3
16	Wall	A proven or honestly-checked LIMIT — the order or result ED does NOT reach (a load-bearing 'no').	Claim	4
17	Synthesis	An arc consolidation / overview / organizing lens; composes standing results, adds no new derivation.	Meta	30
18	Superseded	Retracted / withdrawn / superseded by a later corpus result (kept for provenance).	Meta	3
19	Open	A declared OPEN / unclosed item — a named gap or frontier.	Meta	51